Q12

```c
#include<stdio.h>
void main()
{
    char ch;
    printf("enter any character:");
    scanf("%c", &ch);
    printf("the ASCII value of '%c' is:%d",ch,ch);
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 12file.c -o 12file } ; if ($?)
 { .\12file }
enter any character:A
the ASCII value of 'A' is:65
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q13

```c
#include<stdio.h>
void main()
{
    int a;//where a is any number
    printf("enter a number:");
    scanf("%d", &a);
    if(a%2==0&&a!=0){
        printf("%d is an even number.",a);
    }
    else if(a%2!=0)
    {
        printf("%d ia an odd number.",a);
    }
    else
    {
     printf("%d is zero",a);
    }
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 13file.c -o 13file } ; if ($?)
 { .\13file }
enter a number:55
55 ia an odd number.
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q14

```c
#include<stdio.h>
int main()
{
    char ch;
    printf("enter any value:");
    scanf("%c", &ch);
    if(ch>='A'&& ch<='Z'){
        printf("our inputted character is upper case letter");
    }
    else if(ch>='a'&& ch<='z'){
        printf("our inpuuted character is small case letter");
    }
    else if(ch>='0' && ch<='9'){
        printf("our character is a digit");
    }
    else{
        printf("our character is a special symbol");
    }
    return 0;
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 14file.c -o 14file } ; if ($?)
 { .\14file }
enter any value:&
our character is a special symbol
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q15

```c
#include<stdio.h>
void main()
{
    float maths,chemistry,physics,CS,english,total;
    float percentage;
   printf("enter the marks of physics:");
   scanf("%f",&physics);
   printf("enter the marks of chemistry:");
   scanf("%f",&chemistry);
   printf("enter the marks of maths:");
   scanf("%f",&maths);
   printf("enter the marks of english:");
   scanf("%f",&english);
   printf("enter the marks of computer science:");
   scanf("%f",&CS);
   total=maths+chemistry+physics+CS+english;
   percentage=(total*100)/500;
   printf("total marks=%f\n",total);
   printf("percentages=%f\n",percentage);
   if(percentage>=95){
    printf("grade=A+");
   }
   else if(percentage>=85 && percentage<95){
    printf("grade=A");
   }
   else if(percentage>=70 && percentage<85){
    printf("grade=B");
   }
   else if(percentage>=45 && percentage<70){
    printf("grade=C");
   }
   else{
    printf("student is failed");
   }


}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 15file.c -o 15file } ; if ($?) { .\15file }
enter the marks of physics:88
enter the marks of chemistry:97
enter the marks of maths:80
enter the marks of english:95
enter the marks of computer science:85
total marks=445.000000
percentages=89.000000
grade=A
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q16

```c
#include<stdio.h>
void main()
{
    int num1,num2,num3;
    printf("enter three numbers:");
    scanf("%d %d %d",&num1, &num2, &num3);
    //find max of three numbers
    if(num1>num2){
        if(num1>num3){
            printf("maximum number is:%d", num1);
        }
        else{
            printf("maximum number is:%d",num3);
        }
    }
    else{
        if(num2>num3){
        printf("maximum number is:%d",num2);
        }
        else{
         printf("maximum number is:%d",num3);
        }
    }
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 16file.c -o 16file } ; if ($?) { .\16file }
enter three numbers:55
99
20
maximum number is:99
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q17

```c
#include<stdio.h>
void main()
{
    int num1,num2,num3;
    printf("enter three numbers:");
    scanf("%d %d %d",&num1,&num2,&num3);
    if(num1>=num2&&num1>=num3){
        printf("maximum number is:%d",num1);
    }
    else if(num2>=num3){
        printf("maximum number is:%d",num2);
    }
    else{
        printf("maximum number is:%d",num3);
    }
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 17file.c -o 17file } ; if ($?) { .\17file }
enter three numbers:99
56
100
maximum number is:100
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q18

```c
#include<stdio.h>
void main()
{
    int num1,num2,num3,max;
    printf("enter three numbers:");
    scanf("%d %d %d",&num1,&num2,&num3);
    max=(num1>num2 && num1>num3)? num1 : ((num2>num3)?num2:num3);
    printf("maximum number is:%d",max);
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 18file.c -o 18file } ; if ($?) { .\18file }
enter three numbers:100
1000
500
maximum number is:1000
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q19

```c
#include<stdio.h>
void main()
{
    char ch;
    printf("enter a character:");
    scanf("%c",&ch);
    (ch>='a'&& ch<='z')?printf("'%c' is a small case letter",ch):printf("'%c' is not a small case letter",ch);
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 19file.c -o 19file } ; if ($?) { .\19f
ile }
enter a character:Q
'Q' is not a small case letter
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q20

```c
#include<stdio.h>
void main()
{
    double num1,num2,result;
    char op;
    printf("enter an operator(+,-,*,/):");
    scanf("%c",&op);
    printf("enter two numbers:");
    scanf("%lf %lf",&num1,&num2);
    switch(op){
        case '+':
            result=num1+num2;
            printf("%lf + %lf = %lf",num1,num2,result);
            break;
        case '-':
            result=num1-num2;
            printf("%lf - %lf = %lf",num1,num2,result);
            break;
        case '*':
            result=num1*num2;
            printf("%lf * %lf = %lf",num1,num2,result);
            break;
        case '/':
            if(num2!=0){
                result=num1/num2;
                printf("%lf / %lf = %lf",num1,num2,result);
            }
            else{
                printf("error! division by zero is not allowed.");
            }
            break;
        default:
        printf("error! invalid operator.");
    }
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 20file.c -o
0file } ; if ($?) { .\20file }
enter an operator(+,-,*,/):+
enter two numbers:2
8
2.000000 + 8.000000 = 10.000000
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q21

```c
#include<stdio.h>
void main()
{
    double num1,num2,result;
    char op;
    printf("enter the operator:");
    scanf("%c",&op);
    printf("enter two numbers:");
    scanf("%lf %lf", &num1,&num2);
    if(op=='+'){
        result=num1+num2;
        printf("%lf+%lf=%lf",num1,num2,result);
    }
    else if(op=='-'){
        result=num1-num2;
        printf("%lf-%lf=%lf",num1,num2,result);
    }
    else if(op=='*'){
        result=num1*num2;
        printf("%lf*%lf=%lf",num1,num2,result);
    }
    else if(op=='/'){
        if(num2!=0){
            result=num1/num2;
            printf("%lf/%lf=%lf",num1,num2,result);
        }
        else{
            printf("error! division by zero is not applicable");
        }
    }
    else{
        printf("invalid operator");
    }
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 21file.c -o 21file } ; if ($?) { .\21file }
enter the operator:/
enter two numbers:20
4
20.000000/4.000000=5.000000
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q22

```c
#include<stdio.h>
void main()
{
    int num,factorial;
    printf("enter an integer:");
    scanf("%d",&num);
    factorial=1;
    if(num<0){
        printf("Error! factorial of a negative number does not exists");
    }
    else{
        while(num>0){
            factorial=factorial*num;
            num=num-1;
        }
     }
     printf("the factorial of entered number is:%d",factorial);
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 22file.c -o 22file } ; if ($?) { .\22file }
enter an integer:5
the factorial of entered number is:120
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q23

```c
#include<stdio.h>
void main()
{
    int base,exponent,result,i;
    result=1;
    i=0;
    printf("enter base:");
    scanf("%d",&base);
    printf("enter exponent:");
    scanf("%d",&exponent);
    while(i<exponent){
        result=result*base;
        i=i+1;
    }
    printf("%d raised to the power of %d is:%d",base,exponent,result);
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 23file.c -o 23file } ; if ($?) { .\23file }
enter base:2
enter exponent:5
2 raised to the power of 5 is:32
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```